State Estimation of Induction Motor Using Unscented Kalman Filter

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Abstract—**In this paper, a new estimation technique, unscented Kalman filter (UKF) is applied to state observation in field oriented control (FOC) of induction motor. UKF, a recent derivative-free nonlinear estimation tool, is used for estimating rotor speed and fluxes using sensed stator current and voltages. In the simulations, UKF, whose several intrinsic properties suggest its use over EKF in highly nonlinear systems, turned out to be very similar to EKF in flux estimates. The simulation results also show that UKF has slightly better speed estimation performance than EKF while driven under the identical machine model and parameters (covariances).**

Index Terms—**Induction Motor, Speed Estimation, UKF, EKF, Sensorless Vector Control, State Observer.**

I. INTRODUCTION

In the previous decade, several different methods [1]-[9] were proposed for eliminating speed transducers and Hall effect flux sensors from industrial AC drives in order to reduce the cost and increase the reliability of the overall system. One of the most well-known methods used for speed sensorless field-oriented control of AC drives is Extended Kalman Filter (EKF) [4]-[12]. In the literature, there are several studies which use EKF to estimate rotor speed [4]-[5], machine parameters [6]-[8] and flux [9]. In this paper, rotor speed and flux will be estimated via the model given by Kim and Sul in [5].

Unscented Kalman filter (UKF) is a novel estimation tool introduced by Julier and Uhlmann [13-14] to replace EKF in nonlinear filtering problems. As well-known, EKF is a simple solution derived by direct linearization of the state equation for extending the famous (linear) Kalman filter into nonlinear filtering area. Although it is straightforward and simple, EKF has well-known drawbacks [15]. These drawbacks include

1. Instability due to linearization and erroneous parameters.
2. Costly calculation of Jacobian matrices.
3. Biasedness of its estimates.
4. Lack of analytical methods for suitable selection of model covariances.

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UKF is proposed in order to overcome the first three of these disadvantages. The main advantage of UKF is that it does not use any linearization for calculating the state predictions and covariances. Due to this, its covariance and Kalman gain estimates are more accurate. This accurate gain, at the end, leads to better state estimates. In this study, UKF is introduced into the problem of speed and flux estimation of an induction motor. General simulation results are given and a brief comparison is made between speed estimation performances of UKF and EKF.

In the next section, a more detailed description about the operation of UKF is given. In Section III, the induction motor model configured for UKF is presented. Section IV investigates the simulation results of UKF and makes a comparison between the performance of UKF and that of EKF in terms of induction motor speed estimation errors. Conclusions about our results are summarized in Section V.

II. UNSCENTED KALMAN FILTER

The filtering problem involved in this paper is to find the best (in the sense of minimum mean square error (MMSE)) linear estimate of the state vector x_k of the induction machine which evolves according to the discrete-time nonlinear state transition equation

$$x_{k+1} = f(x_k, u_k) + w_k \tag{1}$$

where $f(.,.)$ is the induction machine dynamics, x_k is the state of the induction machine at sampling instant k, u_k is the known input to the induction machine at time k and w_k is the additive white process noise term representing modeling errors. Also, it is assumed that we have a set of noisy measurements z_k which are related to the state vector of the induction machine by the linear relationship

$$z_k = Hx_k + v_k \tag{2}$$

where H is the properly sized observation matrix and v_k is the white measurement noise related with the measuring device used. The additive white-noise vectors w_k and v_k are Gaussian and uncorrelated from each other with zero mean and covariances Q and R respectively.

The state of the system is assumed to be unknown and therefore; the aim of the estimation process is to find a MMSE estimate of the state $\hat{x}_{k|k}$ which is given by

$$\hat{x}_{k|k} \overset{\Delta}{=} E\{x_k \mid Z^k\} \tag{3}$$

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where $Z^k \overset{\Delta}{=} \{z_1, z_2, ..., z_k\}$ and $E\{x \mid y\}$ denotes the expected value of the quantity x given the information y. Also, traditionally, one calculates the error estimates given by the covariance matrix $P_{k|k}$ defined as

$$P_{k|k} \overset{\Delta}{=} E\{[x_k - \hat{x}_{k|k}][x_k - \hat{x}_{k|k}]^T \mid Z^k\} \tag{4}$$

These direct definitions being too difficult to calculate, recursive forms are adopted for both the state and covariance estimates. The recursive update equations for them are given as

$$\hat{x}_{k+1|k+1} = \hat{x}_{k+1|k} + K_{k+1}\upsilon_{k+1} \tag{5}$$

$$P_{k+1|k+1} = P_{k+1|k} - K_{k+1}P^{\upsilon}_{k+1|k}K^T_{k+1} \tag{6}$$

where the vectors $\hat{x}_{k+1|k}$ (State Prediction), υ_{k+1} (Innovation) and the matrices K_{k+1} (Kalman Gain), $P_{k+1|k}$ (State Prediction Covariance), and $P^{\upsilon}_{k+1|k}$ (Innovation Covariance) are dependent on the quantities $\hat{x}_{k|k}$ and $P_{k|k}$ with the following equations.

$$\hat{x}_{k+1|k} \overset{\Delta}{=} E\{f(x_k, u_k) \mid Z^k\} \tag{7}$$

$$P_{k+1|k} \overset{\Delta}{=} E\{[x_{k+1} - \hat{x}_{k+1|k}][x_{k+1} - \hat{x}_{k+1|k}]^T \mid Z^k\} \tag{8}$$

$$\hat{z}_{k+1|k} = H\hat{x}_{k+1|k} \tag{9}$$

$$\upsilon_{k+1} \overset{\Delta}{=} z_{k+1} - \hat{z}_{k+1|k} \tag{10}$$

$$P^{\upsilon}_{k+1|k} = HP_{k+1|k}H^T + R \tag{11}$$

$$K_{k+1} = P^{xz}_{k+1|k}(P^{\upsilon}_{k+1|k})^{-1} \tag{12}$$

$$P^{xz}_{k+1|k} = P_{k+1|k}H^T \tag{13}$$

The quantities $\hat{x}_{k+1|k}$ and $P_{k+1|k}$, which are called state prediction and prediction covariance of the state respectively, are vital for the overall filter performance. Equations (7) and (8) do not specify how these quantities are calculated. EKF assumes that errors in the state estimates are small enough to approximate (7) and (8) to their first order Taylor series. As a result, $\hat{x}_{k+1|k}$ and $P_{k+1|k}$ are calculated in EKF as follows

$$\hat{x}^{EKF}_{k+1|k} = f(\hat{x}_{k|k}, u_k) \tag{14}$$

$$P^{EKF}_{k+1|k} = \nabla f_x P_{k|k} \nabla f_x^{\,T} + Q \tag{15}$$

where ∇f_x denotes the Jacobian matrix of the function f with respect to the state x. This linearization in EKF frequently yields wrong results in the estimates of the covariance and thus, the state. UKF solves the prediction problem by sampling the distribution of the state in a deterministic manner and then transforming each of the samples using the nonlinear state transition equation.

The n-dimensional random variable x_k with mean $\hat{x}_{k|k}$ and covariance $P_{k|k}$ is approximated by $2n+1$ weighted samples or *sigma points* selected by the algorithm.

$$\chi_0(k \mid k) \overset{\Delta}{=} \hat{x}_{k|k} \quad W_0 \overset{\Delta}{=} \kappa/(n+\kappa) \tag{16}$$

$$\begin{gathered}\chi_i(k \mid k) \overset{\Delta}{=} \hat{x}_{k|k} + (\sqrt{(n+\kappa)(P_{k|k}+Q)})_i \\ W_i \overset{\Delta}{=} 1/(2(n+\kappa))\end{gathered} \tag{17}$$

$$\begin{gathered}\chi_{i+n}(k \mid k) \overset{\Delta}{=} \hat{x}_{k|k} - (\sqrt{(n+\kappa)(P_{k|k}+Q)})_i \\ W_{i+n} \overset{\Delta}{=} 1/(2(n+\kappa))\end{gathered} \tag{18}$$

for $i = 1, \ldots, n$ where $\kappa \in \Re$ is a free real number such that $n+\kappa \neq 0$, $(\sqrt{(n+\kappa)(P_{k|k}+Q)})_i$ is the i^{th} column of the matrix square root of $(n+\kappa)(P_{k|k}+Q)$ and W_i is the weight associated with the i^{th} point.

Given these set of samples, the prediction process is as given below,

1. Each sigma point is transformed through the process dynamics f.

$$\chi_i(k+1 \mid k) = f(\chi_i(k \mid k), u_k) \tag{19}$$

2. The state prediction is computed as

$$\hat{x}_{k+1|k} = \sum_{i=0}^{2n} W_i \chi_i(k+1 \mid k) \tag{20}$$

3. The prediction covariance is calculated as

$$\begin{aligned}P_{k+1|k} = \sum_{i=0}^{2n} W_i[\chi_i(k+1 \mid k) - \hat{x}_{k+1|k}] \\ \times[\chi_i(k+1 \mid k) - \hat{x}_{k+1|k}]^T\end{aligned} \tag{21}$$

The equations (20) and (21) replace (7) and (8). The other UKF operations are the same as (9) to (13). Note that operations in the new set of equations composed by (20), (21), (9) to (13) together with measurement updates given in (5) and (6) use only standard vector and matrix operations and need no derivative and Jacobian approximations. Also, the order of calculation is the same as that of EKF. In the next section, a detailed induction machine model used in the implementation of UKF is given.

III. Induction Motor Model

The dynamic model of the induction motor is given in stationary reference frame which is used in both EKF and UKF for state observation. The rotor speed is regarded to be a parameter where the other state variables of the model are chosen as stator currents and rotor fluxes. The model is given below:

$$\frac{dx(t)}{dt} = Ax(t) + Bu(t) \tag{22}$$

$$y(t) = Cx(t) \tag{23}$$

where

$$x = [i_{ds}, i_{qs}, \phi_{dr}, \phi_{qr}, w_r]^T \tag{24}$$

$$y = [i_{ds}, i_{qs}]^T \tag{25}$$

$$u = [v_{ds}, v_{qs}]^T \tag{26}$$

$$A = \begin{bmatrix} a_{11} & 0 & a_{13} & a_{14} & 0 \\ 0 & a_{22} & a_{23} & a_{24} & 0 \\ \frac{L_m}{T_r} & 0 & -\frac{1}{T_r} & -w_r & 0 \\ 0 & \frac{L_m}{T_r} & w_r & -\frac{1}{T_r} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (27)$$

$$B = \begin{bmatrix} \frac{1}{\sigma L_s} & 0 \\ 0 & \frac{1}{\sigma L_s} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \quad (28)$$

where

$$a_{11} = a_{22} = -\left(\frac{R_s}{\sigma L_s} + \frac{1-\sigma}{\sigma T_r}\right), \quad a_{13} = a_{24} = \frac{L_m}{\sigma L_s L_r T_r} \quad (29)$$

$$a_{14} = -a_{23} = w_r \frac{L_m}{\sigma L_s L_r} \quad (30)$$

The model can be approximately written in the discrete time as

$$\begin{aligned} x_{k+1} &= A_d x_k + B_d u_k \\ y_k &= C_d x_k \end{aligned} \quad (31)$$

where approximations $A_d \approx I + AT$ and $B_d \approx BT$ are done using the assumption that the sampling time T is small.

IV. Simulation Results

To verify the performance of the state estimation, particularly of the speed estimation with UKF, a number of simulations were carried out. The parameters of the motor model are given in the Appendix. In the Figures 1 to 7, the state estimation performance of UKF is simulated and in Figures 8 to 9, accuracies obtained from EKF and UKF are compared for the speed estimation.

Fig. 1 shows the actual state variables of the motor which are stator currents, rotor fluxes and rotor speed at no load in a high speed reversal scheme. Fig. 2 shows corresponding estimated state variables with UKF under the same conditions. There are almost no differences between the actual and the estimated variables. Fig. 3 and Fig. 4 illustrates magnified estimated speed waveforms at no load in four quadrant high speed and low speed reversal schemes respectively. Both the high speed and low speed estimated waveforms confirm that UKF's performance is quite good in speed estimation for all quadrants without causing instability. In Fig. 5, estimated state variables of the induction motor are shown under %100 rated load torque and %100 rated speed conditions. Note that mechanical load is applied to the motor between only 0.75s and 1.5s. In addition to high performance at no load, UKF gives quite satisfactory results under full load condition. In Fig. 6, and Fig. 7, actual and estimated speed characteristics are given on top of each other for %100 and %10 rated torque and speed case. In the transient part of the waveforms, there appears a difference between the estimated and actual values which is the result of the fact that, in induction motor model, the speed is considered as a constant parameter and corrected only in the measurement updates of the UKF. In our trials, we also noticed that there usually exists a small steady state error between the estimated and actual speed values but these seem to be at negligible levels.

In Fig. 8 and 9, the speed estimation performances of UKF and EKF with identical parameters are compared at %100 rated torque and speed. Simulations of Fig. 8 and 9 were carried out for different covariance values. Covariance values were selected so that, in Fig. 8, the steady state performance and in Fig. 9, the transient performance is optimized. It is observed from the figures that, although the performances of the EKF and UKF are close to each other, UKF reduces the transient and steady state speed estimation errors by up to 10 rpm under rated conditions.

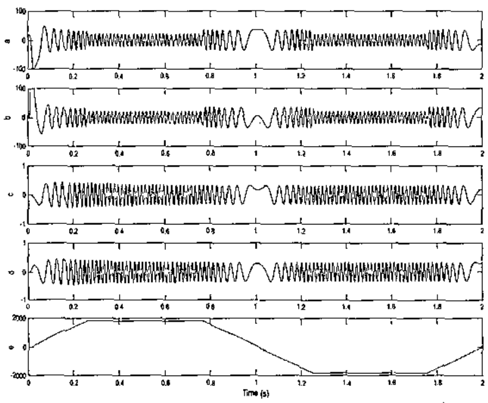

Fig. 1. Induction motor actual states at no load four quadrant high speed reversal (a-b) d-q axis stator currents, (c-d) d-q axis rotor fluxes, (e) rotor speed.

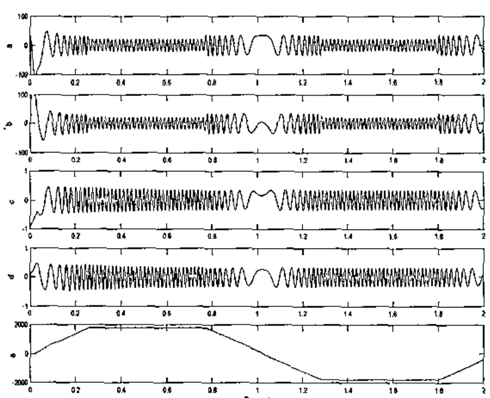

Fig. 2. Induction motor estimated states at no load four quadrant high speed reversal (a-b) estimated d-q axis stator currents, (c-d) estimated d-q axis rotor fluxes, (e) estimated rotor speed.

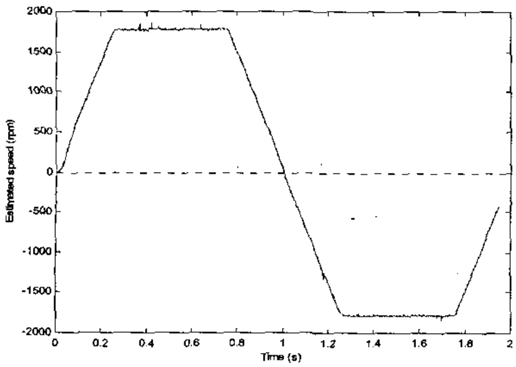

Fig. 3. Induction motor estimated speed at no-load four quadrant high speed reversal.

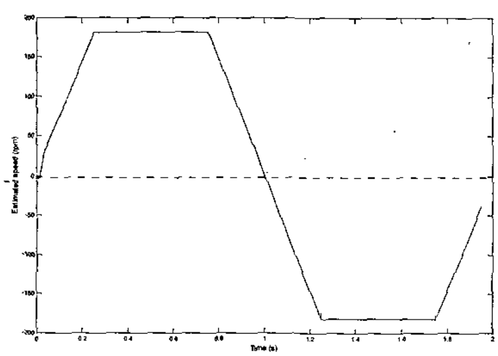

Fig. 4. Induction motor estimated speed at no-load four quadrant low speed reversal.

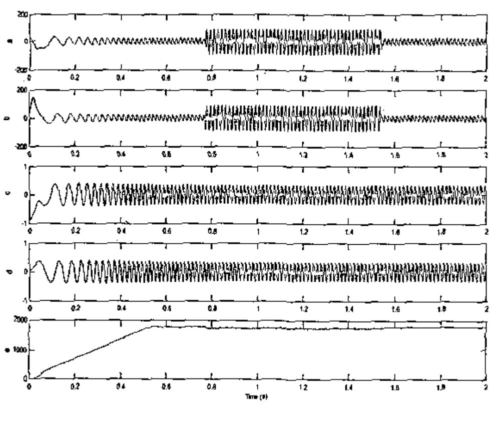

Fig. 5. Induction motor estimated states at %100 rated torque and speed (a-b) estimated d-q axis stator currents, (c-d) estimated d-q axis rotor fluxes, (e) estimated rotor speed (load torque applied between 0.75s-1.5s)

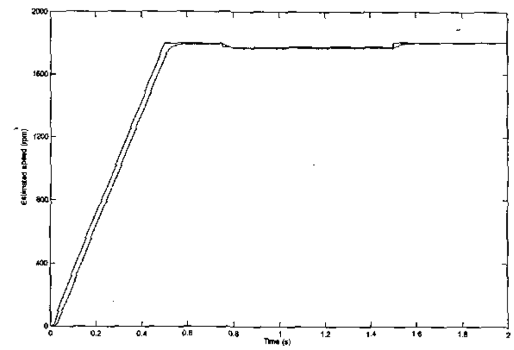

Fig. 6. Induction motor estimated speed at %100 rated torque and speed (load torque applied between 0.75s-1.5s)

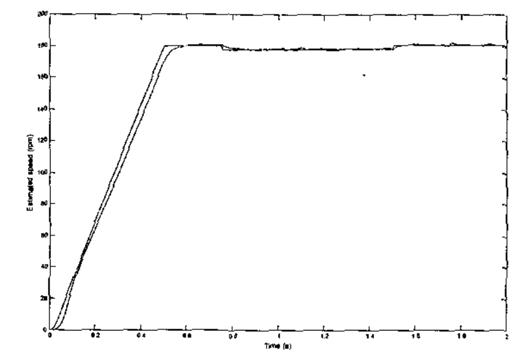

Fig. 7. Induction motor estimated speed at %10 rated torque and speed (load torque applied between 0.75s-1.5s).

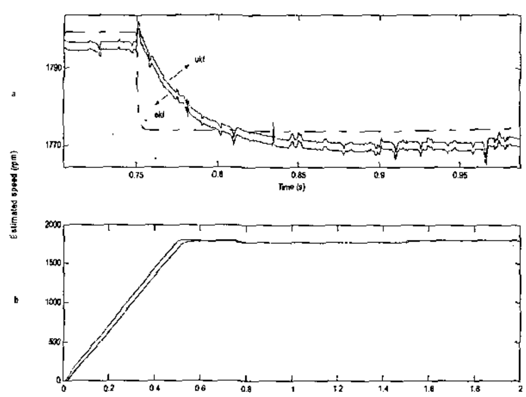

Fig. 8. (b) Induction motor estimated speed optimized for steady state performance at %100 rated torque and speed using EKF and UKF. (a) graphics in (b) zoomed at the mechanical loading initiation. (load torque applied between 0.75s-1.5s)

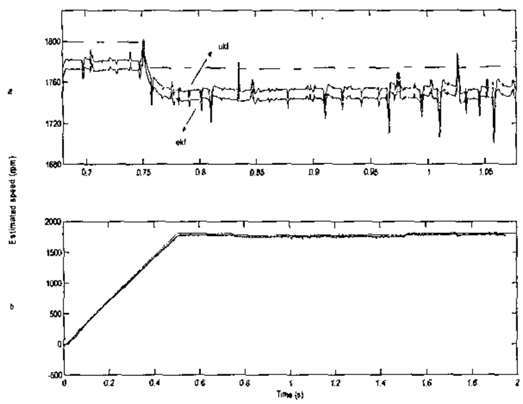

Fig. 9. (b) Induction motor estimated speed optimized for transient performance at %100 rated torque and speed using EKF and UKF. (a) graphics in (b) zoomed at the mechanical loading initiation. (load torque applied between 0.75s-1.5s)

V. Conclusions

The paper adapts a new trend derivative-free nonlinear Kalman filtering technique to the state estimation problem for induction machines. It has been shown that UKF is at least as good as EKF in state observation and it yields even slightly better speed estimation performance than EKF. This result encourages further study in the area to obtain better and better state estimation performances for nonlinear systems to overcome the well-known defects of EKF and other traditional nonlinear filtering techniques.

Nomenclature

i_{ds} :d axis stator current in stationary reference frame

i_{qs} :q axis stator current in stationary reference frame

ϕ_{dr} :d axis rotor flux in stationary reference frame

ϕ_{qr} :q axis rotor flux in stationary reference frame

v_{ds} :d axis stator voltage in stationary reference frame

v_{qs} :q axis stator voltage in stationary reference frame

L_m : mutual inductance

L_s : stator self inductance

T_r : rotor time constant

σ : leakage coefficient factor

Appendix

Rating and parameters of the induction motor used for simulations

20 HP, 220V, 4 poles, 60 Hz

$$R_s = 0.1062\,ohm \qquad L_r = 0.568\,mH$$
$$R_r = 0.0764\,ohm \qquad L_m = 15.5\,mH$$
$$L_s = 0.568\,mH \qquad sigma = 0.0697$$

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